# The Power Law Book

**INDEPENDENT THEMATIC RESEARCH**

**EQUITY STRATEGY · PORTFOLIO INITIATION**

## Quantifying the Physical Bottlenecks of the Global AI CapEx Supercycle

*Overweight Substrate, hold Compute as a counterweight, and size Ballast to absorb drawdown. 23 positions in 5 currencies express a rotation out of hyperscaler beta and into the physical infrastructure that gates it.*

• **The consensus AI trade is crowded.** Hyperscalers and their primary silicon suppliers are the obvious way to own the buildout, and the most efficiently priced. This book is built to express a different, less crowded layer of the same trade.

• **The binding constraint is shifting from silicon to substrate.** Grid interconnect queues and transformer lead times, not chip supply, increasingly gate how fast new capacity can actually come online.

• **Construction is top-down, not stock-by-stock.** Target weights are set at the bucket level (Substrate 42% / Compute 34% / Materials 2% / Ballast 22%); positions inside each bucket are held close to equal-weight so no single stock call carries the thesis.

• **Compute is retained on purpose, not shrunk to a token hedge.** A portfolio that only wins if hyperscalers underperform their own suppliers is a weaker construction than one that wins on the rotation but does not collapse if the rotation stalls.

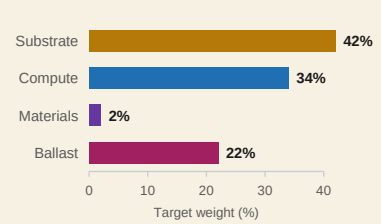
• **Every position carries a standing falsification note:** the specific condition under which its inclusion in the thesis is wrong. The critical-minerals ETF sits in its own Materials bucket rather than being counted as ballast. See Section 4.

### BACKTEST METRICS

| | |
|---|---|
| Index level | **136.69** |
| YTD (USD) | **+44.9%** |
| 1-year (USD) | **+36.7%** |
| Volatility (ann.) | **25.6%** |
| Max drawdown | **–14.7%** |
| Sharpe (vs T-bills) | **1.22** |

*Hindsight backtest, not a track record: today's weights replayed over twelve months, rebalanced monthly, gross of costs.*

### BUCKET ALLOCATION




### AUTHOR

**João**
Business Administration student, ESPM São Paulo
Independent researcher (personal portfolio project)
*Not affiliated with any bank or financial institution*

2. PORTFOLIO CONSTRUCTION

# Bucket Weights and Factor Logic

The book is organized into four thematic buckets rather than picked stock-by-stock. Target weights are set at the bucket level; positions within a bucket are then held close to equal-weight, so no single stock-picking call inside a bucket is doing outsized work. The bet is on the theme, not on any individual name being the best operator in it.

| Bucket | Weight | N | Structural exposure |
|---|---|---|---|
| Substrate | 42.0% | 9 | Fuel, generation, land, interconnect rights and grid equipment: the most direct expression of the thesis |
| Compute | 34.0% | 8 | The hardware layer itself: a counterweight against being wrong about the timing of the rotation |
| Materials | 2.0% | 1 | Rare earths and strategic metals: the export-controlled inputs to magnets, motors and turbines |
| Ballast | 22.0% | 5 | Cross-asset hedges, monetary safety assets, and development-market value, sized to absorb drawdown |

Substrate (42%) is weighted above Compute (34%) deliberately: fuel, land and the grid equipment that connects them are the more direct, less crowded expression of the thesis. Compute is kept at about a third of the book on purpose: a portfolio that only wins if hyperscalers underperform their own suppliers is a weaker construction than one that wins on the rotation but does not collapse if the rotation stalls. Ballast is sized to absorb drawdown from the rest of the book rather than to add a competing return thesis.

Near equal weight is not equal risk. Measured as each line's share of the book's variance (weekly USD returns, 52 weeks), CRWV and NBIS hold 10% of the capital and 31% of the risk; Ballast holds 22% of the capital and 5% of the risk. The full table is on the published page.

## Valuation Discipline and Entry Criteria

A weight target says how much to hold, not what price makes it worth holding. Each bucket carries an explicit initiation threshold and a trim rule, so a position that re-rates away from its thesis is cut on a stated condition rather than on sentiment.

| Bucket | Initiation threshold and trim discipline |
|---|---|
| Substrate | Initiation threshold: Forward EV/EBITDA < 14x or FCF Yield > 5.0%. Position trimmed if NTM EV/EBITDA expands beyond 20x without corresponding reserve/interconnect capacity growth. |
| Compute | Cap total exposure if Basket Forward P/E > 32x or NTM PEG > 1.8. Serves as valuation-disciplined insurance against mistiming the rotation. |
| Materials | Held at the 2% bucket target. Add only when the fund trades at or below NAV; cap the position at five sessions of average traded value on B3. |
| Ballast | Rebalance when yield spread / discount to NAV deviates by >1.5 standard deviations from the 3-year trailing mean. |

### Valuation Snapshot

What each line is priced at, from Yahoo Finance quoteSummary on 2026-10-14. Funds show the fund-level P/E and distribution yield; n.m. marks a negative multiple. * TSM: statements in TWD, quote in USD: statement ratios omitted. * ASML: statements in EUR, quote in USD: statement ratios omitted. Compute basket forward P/E (harmonic mean of TSM, AMD, PLTR, 2357.TW, ASML): **24.6x** against the 32x cap in the Compute rule — inside the cap.

| Line | Fwd P/E | Trail. P/E | EV/EBITDA | FCF yld | Div yld |
|---|---|---|---|---|---|
| URA | — | 21.1x | — | — | 5.2% |
| NLR | — | 19.0x | — | — | 3.1% |
| PETR4.SA | 4.3x | 5.5x | 4.3x | 11.9% | 6.5% |
| 2222.SR | 15.2x | 15.0x | 7.3x | 6.1% | 5.3% |
| LB | 36.4x | 73.4x | 22.5x | 1.8% | 0.5% |
| NBIS | n.m. | 144.9x | 228.1x | -19.1% | 0.0% |
| CRWV | n.m. | — | 23.5x | -25.0% | 0.0% |
| ETN | 25.1x | 41.5x | 28.2x | 2.0% | 1.0% |
| HUBB | 20.5x | 27.9x | 20.1x | 2.0% | 1.2% |
| TSM * | 19.3x | 31.3x | — | — | 0.9% |
| AMD | 36.4x | 145.9x | 102.5x | 0.9% | 0.0% |
| PLTR | 85.3x | 172.9x | 184.9x | 0.4% | 0.0% |
| VGT | — | 32.7x | — | — | 0.4% |
| 2357.TW | 13.1x | 14.6x | 14.2x | -10.3% | 4.3% |
| EWY | — | 9.1x | — | — | 1.1% |
| AINF.L | — | — | — | — | — |
| ASML * | 27.8x | 54.1x | — | — | 0.5% |
| GLD | — | — | — | — | 0.0% |
| AVDV | — | 11.8x | — | — | 3.2% |
| JPM | 13.2x | 14.0x | — | — | 2.0% |
| RARA11.SA | — | 21.8x | — | — | — |
| SGOV | — | — | — | — | 3.7% |
| IEUR | — | 17.4x | — | — | 3.2% |

3. MICROECONOMIC MAPPING

# Position-Level Rationale

## Substrate (42.0%, 9 positions)

- **URA: Global X Uranium ETF · 6.0%**
  Nuclear is the fastest-permitting large-scale answer to hyperscaler 24/7 power demand; the broadest, most liquid way to hold exposure to the fuel cycle.

- **NLR: VanEck Uranium & Nuclear ETF · 5.0%**
  Deliberate overlap with URA on the nuclear theme rather than diversifying away from it: the book accepts that thematic concentration, at an ordinary 5% line, in exchange for broader coverage across the fuel cycle and regulated utilities.

- **PETR4.SA: Petrobras · 6.0%**
  A large, cash-generative conventional producer as a hedge against nuclear permitting delays: thermal generation remains the marginal supplier during interim grid deficits.
- **2222.SR: Saudi Aramco · 6.0%**
  Same logic as PETR4.SA with a different sovereign and cost-curve position; diversifies the conventional-energy leg across geography rather than concentrating it in one producer.
- **LB: LandBridge Company · 5.0%**
  Surface and water rights exposure in the Permian Basin: a direct read on interconnection-queue monetization and data-centre land-siting, the physical layer underneath the generation names.
- **NBIS: Nebius Group · 5.0%**
  Nebius is a compute landlord: capex-funded, customer-concentrated, and competing with the hyperscalers it sells to. Its unit economics are gated by power access and shell space, not by chips.
- **CRWV: CoreWeave · 5.0%**
  The two neoclouds overlap enough operationally that this is closer to one position expressed twice than two independent bets (see Section 4).
- **ETN: Eaton Corporation · 2.0%**
  Owns the electrical equipment layer between the grid and the rack. Its order backlog and lead-time guidance are the most direct listed read on whether the interconnect bottleneck is binding.
- **HUBB: Hubbell Incorporated · 2.0%**
  The utility side of the same bottleneck as Eaton: connecting new load needs transmission and distribution hardware that utilities buy, and that is where its revenue sits.

## Compute & Hardware (34.0%, 8 positions)

- **TSM: Taiwan Semiconductor Manufacturing · 5.0%**
  The central foundry bottleneck and the counterweight the thesis has to survive: near-monopoly on advanced-node and CoWoS packaging capacity captures the demand growth directly if the rotation thesis is wrong or early.
- **AMD: Advanced Micro Devices · 5.0%**
  Second-source accelerator exposure, held at the same weight as its bucket peers rather than sized as a conviction bet: this bucket is insurance against the thesis, not a parallel thesis of its own.
- **PLTR: Palantir Technologies · 5.0%**
  Software/deployment-layer exposure, included so the bucket is not pure hardware and does not miss the demand side entirely.
- **VGT: Vanguard Information Technology ETF · 5.0%**
  A broad, low-conviction sleeve that keeps the bucket from being a handful of single-name bets on a theme this book is intentionally underweighting.
- **2357.TW: ASUSTeK Computer · 4.0%**
  Server assembly exposure: margin-thin and customer-concentrated, and the legacy PC cycle still outweighs the AI server line in reported revenue. Held as a read on physical assembly capacity, not a pure AI-server bet.
- **EWY: iShares MSCI South Korea ETF · 4.0%**
  Korea's memory manufacturing exposure to the buildout, held through the country fund rather than by picking between the two dominant memory makers. It replaced the KOSPI index itself on 14 October 2026 so that every line in the book can actually be bought.
- **AINF.L: AI Infrastructure UCITS ETF (LSE) · 4.0%**
  A packaged, diversified sleeve of infrastructure and connectivity names as a lower-maintenance complement to the single-name bets elsewhere in the bucket.

- **ASML: ASML Holding · 2.0%**
  The single point every advanced fab passes through. Held small because its fate is tied to the same foundry capex cycle as TSM, and the European index line already carries some of it.

### Critical Materials (2.0%, 1 position)

- **RARA11.SA: Rare Earths / Critical Minerals ETF · 2.0%**
  Held in its own Materials bucket rather than in Ballast: a concentrated, policy-driven basket that tracks the same technology-materials cycle as the rest of the book, so in most states it falls with the book rather than against it. The exception is an export-control shock, where it is the one line in the book that re-rates upward. It also traded in only about 24% of the measured window, which limits how much its long-horizon return figures should be trusted.

### Ballast (22.0%, 5 positions)

- **GLD: SPDR Gold Shares · 5.0%**
  Classic drawdown hedge, uncorrelated to the CapEx cycle either way. Held to reduce portfolio volatility, not to express a view.
- **AVDV: Avantis Intl Small Cap Value ETF · 4.0%**
  Deliberately away from the AI theme entirely: developed-market, value-tilted, small-cap exposure as a genuine diversifier rather than a disguised extra bet on the same trade.
- **JPM: JPMorgan Chase & Co. · 4.0% · *credit beta, not ballast***
  A quality financial with real exposure to the debt financing the CapEx cycle runs on. Flagged as credit beta rather than ballast: it is levered to the same economy the rest of the book needs, and the risk table shows it diversifies through low correlation, not through a negative one.
- **SGOV: iShares 0-3 Month Treasury Bond ETF · 5.0%**
  Added after the factor test showed the Ballast still moving with the thesis. Treasury bills carry no equity beta and no AI exposure: dry powder that pays the policy rate while the rest of the book is marked down.
- **IEUR: iShares Core MSCI Europe ETF · 4.0%**
  Chosen over the STOXX Europe 600 fund (EXSA) because it holds less ASML: 4.34% against 4.41% in the issuer's holdings files for 13 Oct 2026. It tracks MSCI Europe IMI, not the STOXX index.

4. ANALYTICAL DISCLOSURES

# Known Portfolio Weaknesses

Every position in the live dashboard carries a standing falsification note: the specific condition under which its place in the thesis is wrong, refreshed alongside the daily data pull. The intent is to make it structurally harder to only look for evidence that confirms the thesis. The open weaknesses, stated plainly:

**Concentration.** Measured on weekly returns, the 23-position book behaves like 1.8 effective bets, with 85% of its variance on one common factor.

**Materials concentration and liquidity drag.** RARA11.SA sits in its own Materials bucket rather than in Ballast, because it is correlated with the same technology-materials cycle the rest of the book is long. Its thin trading history (~24% active coverage in the measured window) adds noise to any long-horizon variance figure.

**Temporal horizon mismatch.** Daily deviation tracking compares short-term market noise against a multi-year infrastructure lead-time thesis (permitting and transformer procurement run three to five years). Daily relative performance between Substrate and Compute is an operational monitoring routine, not statistical validation.

### Near-Term Monitoring Catalysts

The mismatch above is real but not unanswerable. Three things resolve inside nought to four quarters and bear directly on whether the physical constraint is binding, which is what daily price action cannot tell you:

- **Policy & Grid Reform.** Track compliance timelines and cluster-study clearing under FERC Order 2023 (US interconnection queue reform), alongside PJM and ERCOT capacity auction clearing prices. Queue throughput is the single most direct read on whether the interconnect bottleneck is binding or easing.
- **Supply Chain Lead Times.** Monitor quarterly management guidance on high-voltage transformer and switchgear lead times from key equipment OEMs — Eaton Corp (ETN), Hubbell (HUBB) and Schneider Electric. Guidance revisions front-run the physical constraint by several quarters.
- **Nuclear PPAs & Co-location.** Track announcements of direct-connect nuclear Power Purchase Agreements (Constellation Energy, Talen Energy, Vistra) and hyperscaler co-location approvals. Each signed PPA is a datable observation that baseload scarcity is being priced.

## Variant View and Falsifiers

The market prices the AI buildout as a semiconductor story and treats power as a utility input that will be supplied on demand. This book's view is that grid interconnection, fuel and sited land are the binding constraint for the next three to five years, so the scarce rent accrues to whoever owns power, fuel and land with grid access rather than to the designers of the chips; the consensus reprices that only once announced data-centre capacity visibly slips.

**Horizon.** Three to five years, matching transformer, switchgear and interconnection-queue lead times. Daily and monthly moves are monitored, not acted on. **Review and exit.** Reviewed each quarter against the portfolio falsifiers below and every line's own 'what breaks it'. A line is cut when its falsification condition is met, not on price alone; a line whose valuation breaches its bucket's trim rule is cut back to target. If two portfolio falsifiers trigger in the same quarter, the thesis itself is reconsidered, not just the lines.

| Falsifier | Test | Reading | Status |
|---|---|---|---|
| Hyperscaler capex cut | Two or more of Microsoft, Alphabet, Amazon and Meta cut next-year capital spending guidance in the same quarter. | checked each earnings season | watch |
| Compute keeps leading power | Compute out-returns Substrate by more than 20 percentage points over twelve months. | +56.7pp | **triggered** |
| Uranium price breaks | Sprott Physical Uranium Trust (a holder of physical uranium) falls more than 25% over twelve months. | -0.9% | holding |
| Memory pricing rolls over | Micron, the listed read on HBM pricing, trades more than 30% below its twelve-month high. | -21.4% | holding |
| Oil floor gives way | Brent trades below $60, the level at which Saudi Aramco and Petrobras payouts come under pressure. | $105.69 | holding |

**23 lines, 1.8 effective bets.** Splitting the book's weekly variance across its principal components, the first carries 85% of it; the entropy-based effective number of bets is 1.8 of 22 lines. Shared points of failure: Taiwan 9% of capital, 9% of risk; Korea memory 4% of capital, 7% of risk; US AI capex 29% of capital, 55% of risk; Nuclear fuel 11% of capital, 16% of risk.

## Change Log

The figures in this note replay today's book, so every change below rewrites them; the live record on the published page is chain-linked from its inception and is not rewritten. Entries before 14 October 2026 are reconstructed from the repository history.

| Date | What changed | Why |
|---|---|---|
| 2026-10-14 | RARA11.SA moved from Ballast to a new Materials bucket (5%); Ballast now 25%. | A rare-earth basket falls with the book rather than against it, so it was never ballast. |
| 2026-10-14 | EWY replaced ^KS11 at 4% in Compute. | Every line must be investable. The KOSPI is kept as EWY's reference index. |

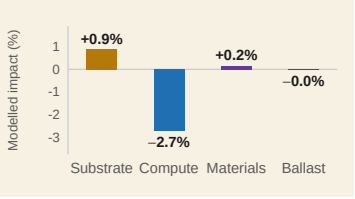
| Date | What changed | Why |
|---|---|---|
| 2026-10-14 | Headline rebalanced monthly instead of daily; volatility annualised at each series' own observation frequency. | Daily rebalancing harvests a premium no real book collects at zero cost; the union-calendar curve has ~309 observations a year, not 252. |
| 2026-10-14 | Bars keyed by each exchange's own trading date; hedge carry from dated USD and BRL rates; three-factor, three-year regressions on excess returns. | FX bars arrive stamped the day before, so every non-USD close had been converted at the next day's rate; an independent recomputation of PETR4.SA found it. |
| 2026-10-14 | Substrate (renamed from Energy) 42: adds ETN 2 and HUBB 2. Compute 34: adds ASML 2. Materials 2: RARA11.SA 5 -> 2. Ballast 22: AVDV 5 -> 4, JPM 5 -> 4, IEUR 4 added, XLP removed. | Grid equipment is the bottleneck the thesis names, so it belongs in the bucket beside fuel and land; ASML is the one supplier every fab depends on. IEUR (MSCI Europe) replaces XLP as Ballast's equity line, chosen over EXSA for its lower ASML weight. RARA11 is cut to an option-sized line. **What breaks it:** a rotation out of AI into defensives, which XLP caught in the book's worst stretch (+7.5%, 2 Jun - 29 Jul 2026), now lands only on SGOV and GLD, and Ballast loses its one equity line that rose while the thesis fell. |
| 2026-10-08 | Index switched from weight × price to compounded weight-averaged daily returns; series switched to dividend-adjusted (total-return) closes. | The price-weighted version let a $400 GLD share outvote a $5 KOSPI point regardless of weight; price-only closes read a T-bill fund as flat. |
| 2026-10-08 | SGOV and XLP added to Ballast at 5% each; GLD, AVDV, JPM and RARA11.SA cut to 5% so Ballast stayed at 30% (18 → 20 positions). | The factor test showed Ballast still moving with the thesis. XLP rose 7.5% in the book's worst stretch (2 Jun – 29 Jul 2026); SGOV has had no meaningful drawdown. |
| 2026-08-12 | Book rebuilt as 18 positions in three buckets (Energy 38 / Compute 32 / Ballast 30), equal weight within each bucket. Added NBIS, CRWV, AMD, TSM, VGT, RARA11.SA, GLD and JPM; dropped a privately marked holding; replaced a dated gold future with GLD. | Every line publicly priced; spot-backed gold shares remove the roll and expiry. |

5. RISK FRAMEWORK

# Scenario Stress-Testing

An empirical covariance matrix, estimated over a year of daily data across five currencies (one line with 12 weeks of trading history), would look statistically rigorous while being almost entirely noise. The portfolio uses stated-assumption scenario shocks instead.

> Interconnection queues and transformer lead times, not chip supply, become the binding constraint; announced data-centre builds slip. Modelled portfolio impact **–1.7%** (Substrate +0.9%, Compute –2.7%, Materials +0.2%, Ballast –0.0%). This is the scenario the book is built to profit from being right about: the one where a negative headline number still confirms the thesis is working as designed.




**CapEx Retrenchment** (cyclical digestion): hyperscalers cut AI capital-spending guidance and the supercycle is re-rated as an ordinary cycle. This is the scenario the Compute bucket exists to cushion against, and the one most likely to hurt every bucket at once. Modelled impact **–17.9%** on the stated shocks; the position sizing, not a forecast, is the hedge.

**Geopolitical Shock** (Taiwan Strait escalation): advanced foundry output is interrupted and export controls tighten on strategic materials. The tail scenario the portfolio is least hedged against: TSM, 2357.TW and EWY all sit in the impact zone and are closer to one bet than three. RARA11.SA is the one line that works here, since the same export controls that interrupt foundry output bid rare earths up, but at 2% it cannot offset a Compute bucket 17 times its size. Modelled impact **–10.3%** on the stated shocks, which treat the three Taiwan- and Korea-linked lines as impaired rather than merely marked down.

### The Historical Record

Today's weights held through the worst stretches of the last five years, using each line's own history where it existed and a stated stand-in where it did not (NBIS as SMH, CRWV as SMH, LB as TPL, RARA11.SA as REMX, AINF.L as VGT). Historical one-day VaR at 95% is 2.5% (CVaR 3.2%), one-week VaR at 95% 4.4% (CVaR 5.8%).

| Scenario | Window | Book | VGT | ACWI | 60/40 |
|---|---|---|---|---|---|
| 2022 bear market | 2022-01-03 to 2022-10-12 | **–17.3%** | –34.5% | –26.4% | –21.6% |
| Rate spike, 10-year to 5% | 2023-07-31 to 2023-10-19 | **–1.9%** | –8.1% | –7.9% | –6.9% |
| DeepSeek day | 2025-01-24 to 2025-01-27 | **–6.9%** | –5.2% | –1.2% | –0.5% |
| Tariff shock, oil to ~$60 | 2025-04-02 to 2025-04-08 | **–13.3%** | –14.7% | –11.8% | –7.5% |
| Tech falls 40% (dot-com style) | beta 0.76 to VGT | **–30.5%** | –40.0% | — | — |

### Exposure, Not Alpha

Weekly excess returns over 2023–2026 load ACWI 0.64, SMH 0.30, XLE 0.07 on global equity, semiconductors and energy, with $R^2$ 0.72. Returns are explained by factor exposure; an intercept in a backtest of positions chosen with hindsight measures the selection, not an edge. Over the latest year the book's Sharpe ratio was 1.22, against 1.24 for VGT (US tech) and 1.03 for ACWI (global equity): it did not beat simply holding tech per unit of risk.

> **Why stated assumptions instead of a fitted model.** A covariance matrix estimated from a year of daily data across five currencies would look statistically rigorous and would be almost entirely noise, given how little history several positions have. The three scenarios are therefore stated shocks, written down per position in data/scenarios.json so they can be argued with, and checked against what actually happened in the historical windows above rather than presented as forecasts.

6. TECHNICAL STACK

## Ingestion Pipeline & Architecture

The dashboard is built and deployed on Streamlit. Text generation for the daily note and the per-position falsification commentary runs through a deterministic, rule-based renderer rather than a live LLM call. No API key is configured in the deployed version, so the analytical judgment embedded in the classifications and the self-critique is fixed at build time, not generated fresh each day. What refreshes on weekdays is the price, volume, and news data feeding that fixed framework: Google News RSS, SEC EDGAR's Atom feed for filings, and the Yahoo Finance chart v8 endpoint for prices. This is a deliberate choice worth stating plainly, since it affects what the analytical work in the project actually is: it sits in the research and portfolio-construction decisions, not in a model narrating markets live.

## IMPORTANT DISCLOSURES

This document is prepared solely by the author as a personal research and portfolio-construction exercise, intended to demonstrate analytical process for professional evaluation purposes such as a CV or job application. It is not prepared by, affiliated with, reviewed by, or endorsed by any bank, broker-dealer, or financial institution, and its formatting is modelled on sell-side research conventions for presentation purposes only. It does not constitute financial advice, an offer to buy or sell any security, or an investment recommendation, and the author is not a licensed financial advisor. Market data is delayed and sourced from free public endpoints. Performance figures are a backward-looking simulation of current portfolio weights over a window ending on the report date and do not represent a live track record. The thesis, positions and weights are the author's; all software behind this report, the data pipeline and the published page was written by Claude Code, Anthropic's AI coding agent, under the author's direction. An educational exercise, not a securities analysis report: the author is not a CVM-registered securities analyst (Resolução CVM 20) or investment adviser, and nothing here is a recommendation.